GCI WORLD 2026 · FINAL ASSIGNMENT

Profit-Driven Customer Retention

Predicting which telecom customers will leave, and deciding who should get a retention offer

Nishank Chandoliya

Client: Company A (anonymous telecom operator) 100,000 customers XGBoost · SHAP · profit-based threshold

SUMMARY

Who will leave, and who is worth an offer

THE PROBLEM

Company A pays to replace every customer who leaves. Who is likely to leave, and who is worth an offer?

WHAT I BUILT

An XGBoost churn score for every customer (ROC-AUC about 0.70), SHAP to explain it, and a profit-based rule for who gets an offer.

WHAT I FOUND

Older devices mean more churn: 40% under 90 days vs 60% after 3 years. Churn peaks at 13–18 months of tenure.

WHAT I PROPOSE

Score customers monthly, contact the highest-risk ones first with an offer that fits their main driver, and pilot it before rollout.

MARKET CONTEXT

Growing market, moving customers

$2.1T

Global telecom services market in 2025, expected to reach $2.2T in 2026 [1]

21.5%

Average yearly customer churn in the telecom industry [2]

$400

Average cost to win a new wireless subscriber, about 5× the cost of keeping one (US, 1998) [3]

Every customer who leaves has to be replaced at a high cost. Keeping existing customers is the cheaper way to grow.

[1] Grand View Research · [2] Focus Digital, 2026 · [3] Mozer et al., 1999

THE DATA

100,000 customers, two tables, 100 columns

Client.csv
50 columns: account, credit class, handset, household data

Record.csv
51 columns: usage, billing, call quality, tenure, churn

Joined table
One row per customer, joined on Customer_ID

49.6%
of customers churned. Balanced on purpose: a big US operator loses about 0.9% a month [4]

FEATURE GROUPS

- Revenue and billing
- Usage and activity
- Usage and revenue trends
- Network quality
- Voice and data calls
- Customer care
- Roaming
- Handset and device age
- Tenure
- Household and demographics
- Geography
- Lifestyle segments

Data: GCI World 2026 course, Company A · [4] AT&T postpaid phone churn, Q3 2025 (PhoneArena)

EXPLORATORY ANALYSIS

Three hypotheses, checked against the data

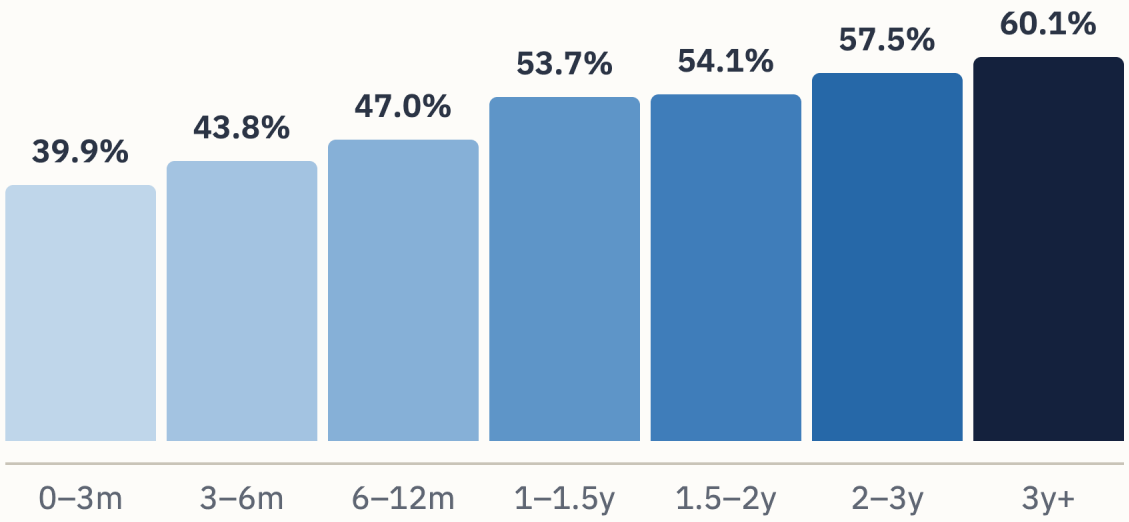
1	**Older devices drive churn**	✓ Supported	Churn rises from 39.9% (device under 90 days old) to 60.1% (over 3 years). Device age is also the #1 SHAP driver.
2	**New customers are the most fragile**	⚠ Not supported	6–12 month customers churn below average (46.8% vs 49.6%). Churn peaks at 13–18 months (52.7%), around when a first-year deal may end.
3	**Usage trend matters more than usage level**	🔍 Partly	Both matter. A usage trend (change_mou) is the #3 SHAP driver, usage level is #4, and the new 3-vs-6-month ratio is #8.

Churn rates from the full dataset (100,000 customers); SHAP ranks from the model on slide 12

EXPLORATORY ANALYSIS

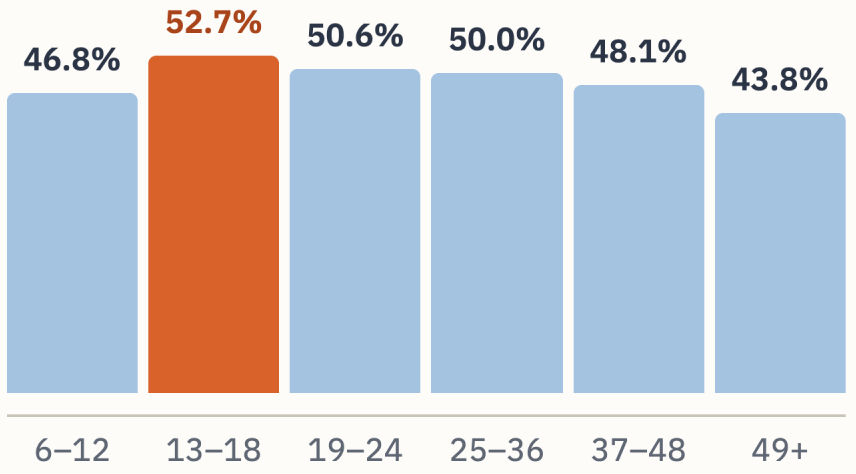
Churn by device age and by tenure

Churn rate by device age



Every step up in device age (m = months, y = years) means more churn. 133 rows with a negative age were left out.

Churn rate by tenure (months)



Highest at 13–18 months, lowest for the longest customers. Average for all: 49.6%.

All 100,000 customers; bar heights start at 0%

FEATURE ENGINEERING

Churners use less and their usage is flat

Usage level: minutes per month (median)



Churners use about 13% fewer minutes. Heavy users do not churn more.

Usage trend: last 3 months ÷ last 6 months (median)

1.018
Retained: still growing

1.002
Churned: flat

1.0 means no change. The gap is small but points the same way as the usage-change column.

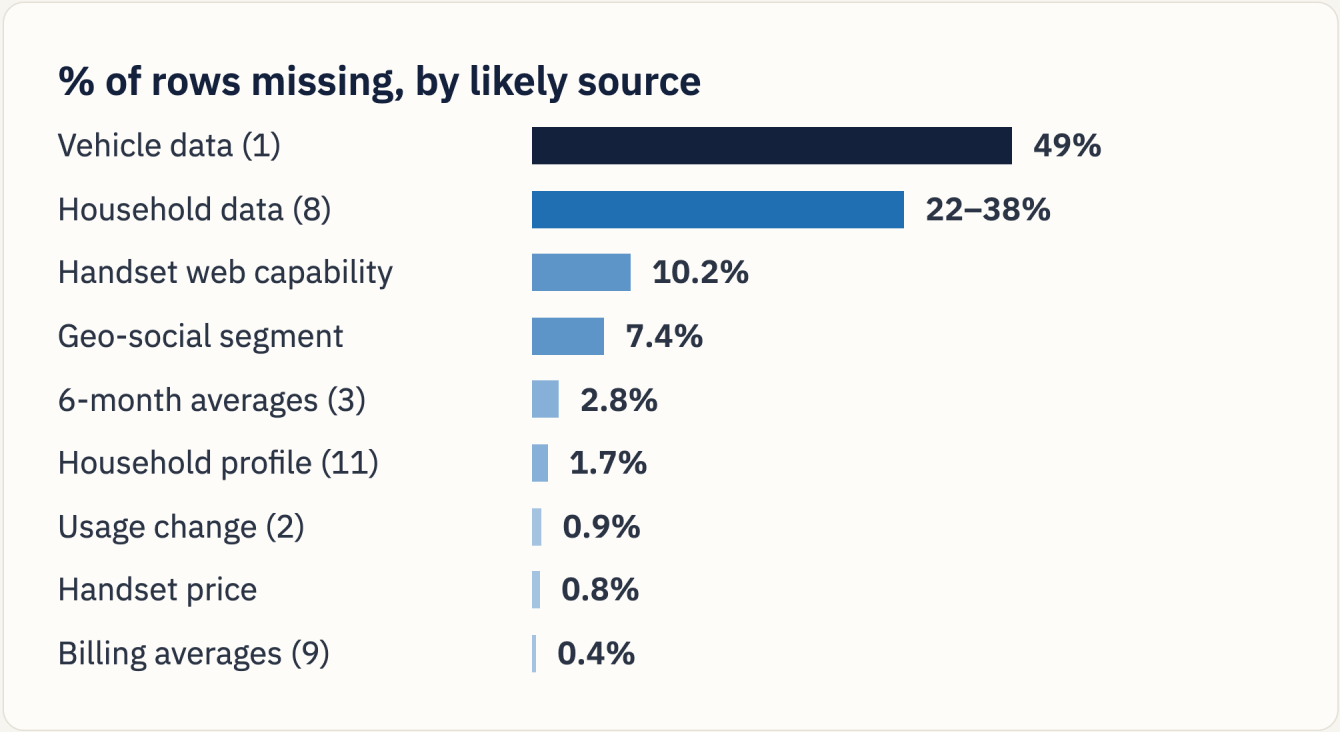
NEW FEATURES

usage_momentum_ratio = avg3mou ÷ avg6mou, and revenue_momentum_ratio = avg3rev ÷ avg6rev (a tiny number is added to avoid dividing by zero)

Medians over all 100,000 customers, before filling missing values

DATA QUALITY

Missing values come in groups



How they were filled

6-month averages: accounts under 6 months use their 3-month average, the rest the median

Usage change: set to 0, there is no earlier period

Geo-social segment: kept as its own "Missing" group

Everything else: median for numbers, "Unknown" for text

Columns missing in exactly the same rows likely come from the same source system. The source names are my guess.

43 of 100 columns have missing values; no rows were dropped

PROBLEM DEFINITION

Predict who will churn, then pick who to contact by profit, not accuracy

BUSINESS OBJECTIVE

Give the retention team a ranked, explained list of customers where contacting them is expected to pay off.

ML TASK

Binary classification with a churn probability, so the cut-off can follow budget or profit instead of a fixed 0.5.

TARGET

churn (1 = left 31–60 days later). A clear label tied to one action with a clear cost per outcome.

Caveat: the 49.6% churn rate in this data is balanced on purpose. Real churn is far lower, so dollar results must be rescaled before real use.

MODEL

XGBoost on 20,000 held-out customers

ROC-AUC (MAIN)	ACCURACY	RECALL (CHURN)
0.6952	0.6394	65%

Confusion matrix at the default 0.5 threshold · precision for churn: 63%

	Predicted: stays	Predicted: churns
Actually stayed	6,359 correct	3,729 false alarms
Actually churned	3,483 missed	6,429 caught

Why these choices

XGBoost handles about 100 mixed features and non-linear effects; row and column sampling limits overfitting.

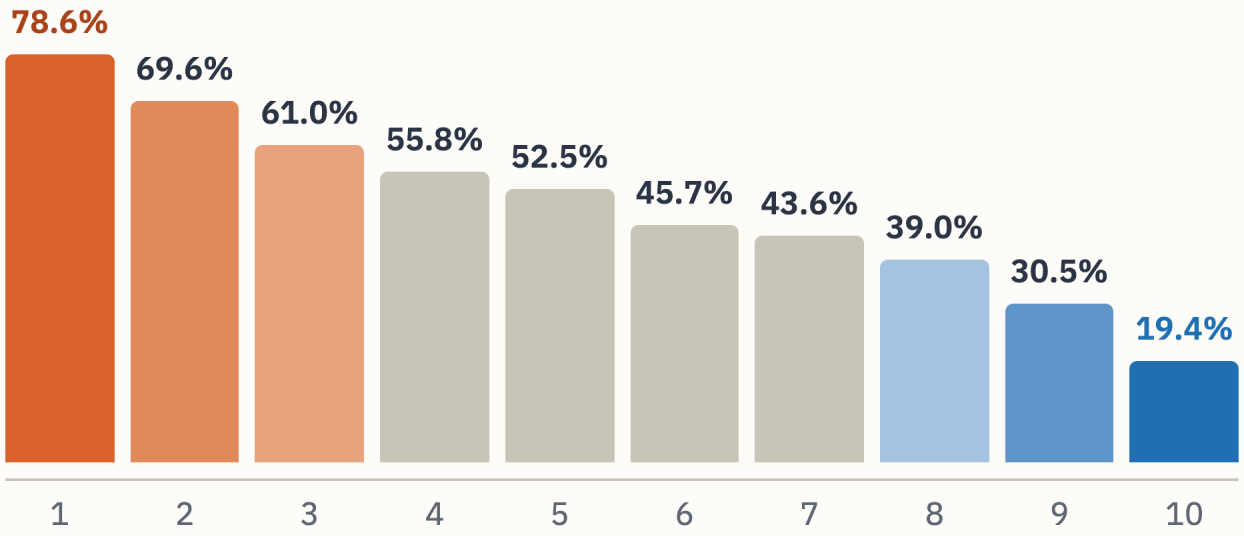
ROC-AUC measures how well customers are ranked at every threshold. The real threshold is picked later, by profit.

Setup: 350 trees, depth 6, learning rate 0.05, stratified 80/20 split, fixed seed.

Test set: 20,000 customers (9,912 churned) · later: 5-fold CV 0.696 ± 0.004, with extra features 0.703

MODEL

The score ranks customers well



78.6%
churned, top 10%

19.4%
churned, bottom 10%

When the budget covers only part of the base, start from the top of this list.

No single column is strongly linked to churn: the largest correlation is 0.11 (device age). The score works by combining many weak signals.

20,000 test customers, 2,000 per group

EXPLAINING THE MODEL (SHAP)

Device age and tenure drive churn the most

Feature	Average impact on the score	Direction
Device age · eqpdays	0.25	Older device → higher risk
Tenure · months	0.17	Highest at 11–15 months
Change in usage · change_mou	0.15	Falling usage → higher risk
Usage level · mou_Mean	0.11	Lower usage → higher risk
Handset price · hnd_price	0.09	Cheaper handset → higher risk
Monthly plan charge · totmrc_Mean	0.09	Lower charge → mostly higher risk
Subscribers · uniqsubs	0.09	More subscribers → higher risk
Usage trend ratio · new feature	0.08	Falling trend → higher risk
Dropped calls · drop_vce_Mean	0.08	More dropped calls → higher risk

Mean |SHAP value| on 1,000 random test customers · direction read from the SHAP values

EXPLAINING THE MODEL (SHAP)

One customer: a score, a reason and an action

HIGHEST RISK IN THE SAMPLE

92.1%

predicted churn probability

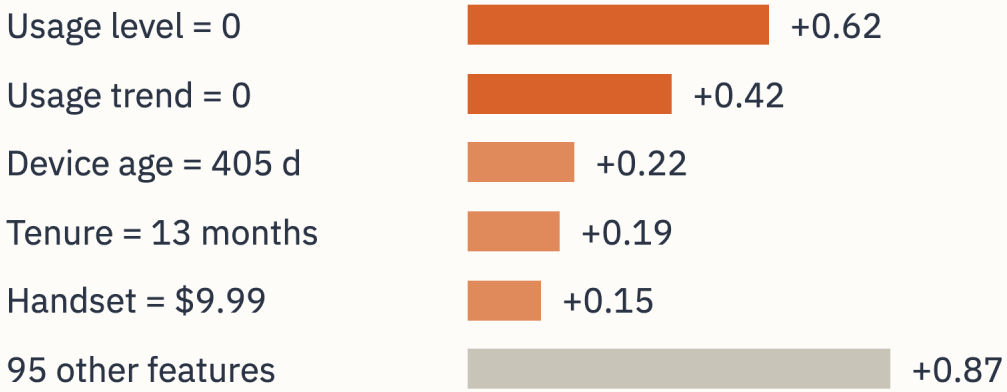
0 minutes of use in the last 6 months

Device 405 days old

13 months with Company A

$9.99 handset, $10 monthly plan

What pushed the score up (SHAP, log-odds)



Average score −0.02 → this customer 2.45, which is 92.1%

Action: device upgrade offer. Zero usage, so first check the line is still in use.

Highest churn probability among the 1,000 test customers used for SHAP

FROM SCORE TO DECISION

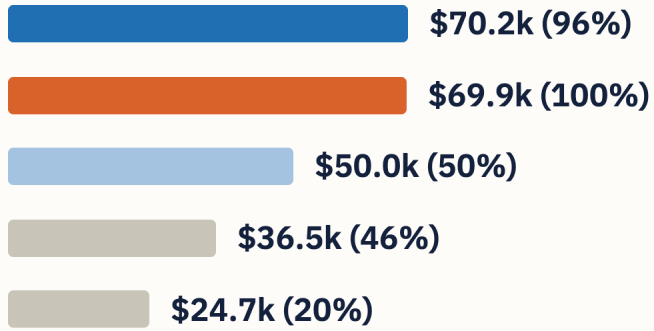
Count each customer once, with a success rate

Value of one decision

Customer	Gets an offer	No offer
Would churn	s × $705 − $35	$0
Would stay	−$35	$0

$705 = $58.77 average monthly revenue × 12 months. s = success rate, the share of offered churners who stay. It is unknown; 30% is the middle case. Profit is counted against doing nothing.

Profit per 1,000 customers at 30% success



At 10% success the model still makes $5.0k per 1,000 customers while offering everyone loses $49. Only then does the ranking clearly pay.

RESULT With half the customers churning, the best rule is almost "offer everyone".

20,000 test customers · threshold picked on separate training data · (%) = share who get an offer

REALITY CHECK

At a real churn rate, a $35 offer rarely pays

−$30.8k

per 1,000 customers if everyone gets the offer (2% churn, 30% success)

4.6%

of customers the model would contact even at 100% success; almost none at 30%

$17–$56

most an offer can cost and still pay for the top 10% of scores (30% to 100% success)

What this means

The model ranks customers well (slide 11), but the real churn rate, offer cost and success rate decide if a campaign makes money.

Cheap offers for most customers; keep costly ones, such as calls or device vouchers, for the top group.

Measure the success rate in a pilot before setting the budget.

Test customers re-weighted to 2% churn · real monthly churn: 2–3% [3], 0.92% at AT&T in Q3 2025 [4]

PROPOSAL

Score, rank, explain, act: every month

1 · Score
Every customer, every month

→

2 · Rank
Start from the top, within budget

→

3 · Explain
Top SHAP driver per customer

→

4 · Act
Offer that matches the driver

Main driver → suggested action

Main driver	Action
Old device (eqpdays)	Device upgrade offer, e.g. zero-down upgrade with a new contract
Tenure close to 12 months	Reach out at months 10–12, before churn peaks
Falling usage (change_mou)	Re-engagement offer: data bonus or loyalty reward
Dropped calls (drop_vce_Mean)	Proactive call with a billing credit, and check the network
High overage charges (ovrrev_Mean)	Move to a plan that fits, to stop surprise bills

Action mapping as used in the notebook; timing from the tenure analysis (slide 6)

NEXT STEPS

Ready for a pilot, not for full rollout

Risks

- **Data:** trained on a balanced research file; results must be checked on live data first.
- **Capacity:** a low threshold flags almost everyone; use tiers, such as automated offers for medium risk and calls for the highest risk.
- **Fairness:** the data has demographic fields such as ethnicity; check for bias before using scores to give different offers.
- **Drift:** competitor moves change why people leave; retrain and reset the threshold every quarter.

Rollout plan

PHASE 1 · 0–3 MONTHS

Retrain on live data with real costs; A/B pilot on flagged customers.

PHASE 2 · 3–6 MONTHS

Scores and reasons in the CRM; driver-to-action playbook.

PHASE 3 · 6–12 MONTHS

Monthly scoring for all; compare real profit with the forecast.

SOURCES

References

[1] Grand View Research. Global Telecom Services Market: size and outlook. grandviewresearch.com/industry-analysis/global-telecom-services-market

[2] Focus Digital (C. McGee, 2026). Average Churn Rate by Industry: 2025 Benchmarks. focus-digital.co/average-churn-rate-by-industry

[3] Mozer, M. C., Wolniewicz, R., Grimes, D. B., Johnson, E., Kaushansky, H. (1999). Churn Reduction in the Wireless Industry. NIPS 12. papers.nips.cc

[4] PhoneArena. “The numbers don’t lie and AT&T is last among Big Three” (AT&T postpaid phone churn, Q3 2025). phonearena.com

[5] Chen, T., Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD 2016.

[6] Lundberg, S. M., Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS 2017.

Data: “Company A” telecom dataset, GCI World 2026 course (Matsuo-Iwasawa Lab, University of Tokyo). Not public.